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The Future of Insurance Starts Here

Responsive Page Design Guide





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1 Introduction

This document serves as Digital 1st platform's guide for developing Responsive Web Applications. Note that this is not a full guide/documentation for Digital 1st platform.

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2 Before App Development

2.1 Spacing between elements

To ease development of application pages/forms, Page designer in Digital 1st platform provides default vertical spacing between elements. This can be changed through theme functionality of the platform. Please set following theme variables with appropriate values.

1. top-margin-for-widget-inside-column
2. bottom-margin-for-widget-inside-column

Below figures shows the difference. Note the spacing on top, bottom and between "First Name" and "Last Name" fields.



Figure 1: With default spacing between elements



Figure 2: With no spacing between elements

- *If value is too low, one may need to provide top/bottom margins to add some space. If values is high then one may need to provide negative top/bottom margin to bring elements nearer.*
- *These variables can be changed any time. However, it is **strongly** recommended to adjust these values before developing any pages, as it will affect already created pages.*

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3 Units

Digital 1st Insurance platform allows following units for most of the Properties (provided it's applicable) and theme variable values.

- px (Pixels*)
- % (Parentage)
- em (Relative to the font size of the element)
- rem (Relative to the font size of the root element)
- vh (Relative to 1% of the width of the viewport**)
- vw (Relative to 1% of the height of the viewport**)

- Pixels (px) are relative to the viewing device. For low-dpi devices, 1px is one device pixel (dot) of the display. For high resolution screens 1px implies multiple device pixels.
- **Viewport equals the browser window size. If the view port is 50cm wide, 1vw = 0.5cm.



Figure 3: Units for Font Size

- *Usually em and rem are practical for scalable layouts.*

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4 Default Responsiveness

4.1 Bootstrap at core

Digital 1st platform uses Bootstrap 3.x library to enable responsiveness. Page design follows bootstrap grid structure (Rows and Columns) to organize page elements. Hence, elements wrap vertically, by default, on mobile view.



Figure 4: Desktop view of sample form on page

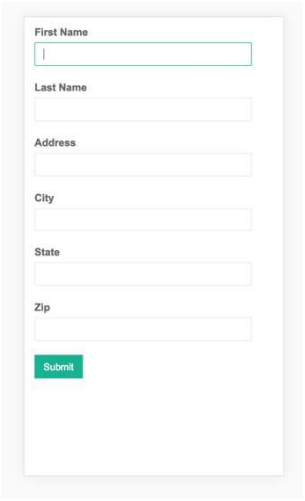


Figure 5: Mobile view of sample form on page

- *To control width of each column in a given row, "Col-span" property can be used (allows col-span of 1 to 12). Note that col-span is not applied in mobile view unless "Wrap content on small screen" Property is set to "No".*

4.2 Flexible Rows

Rows/Inner-Rows can be marked as Flex row. Flex row arranges content one after another. One can use "Space Between" property to specify, as name suggests, space between content. It also supports following Properties.

1. Justify Content: Controls horizontal alignment of content.

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2. Align Items: Controls vertical alignment of content.
3. Flex Wrap: Controls wrapping behavior for content.

- *"Col-span" property is not applicable for columns inside Flexible Rows.*

4.3 Background Images and Color

4.3.1 Background Cover

If single background image is used for all devices then set following property as suggested, unless different behavior is required. This will ensure that image does not appear stretched/squeezed on different devices.

- Background Position: Center Center
- Background Repeat: No Repeat
- Background size: Cover

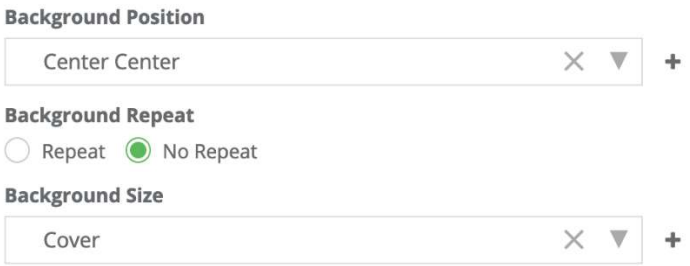


Figure 6: Mobile view of sample form on page

4.3.2 Offset for Row/Column

When background image or Color is applied to Row/Inner-Row (non-flex) or Column and it needs to stretch from end-to-end on screen; one can set following Property to Yes.

Property:

- Offset Row/Column Margin For Background Image/Color

Example:

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Screen having Header, Mid-section and Footer; created using three Rows with either background image or Color. Set above-mentioned property to stretch the background image or Color end-to-end of screen.

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5 Advanced Controls

Over and above default responsiveness, Digital 1st Insurance offers advanced controls and capabilities to further achieve desired page responsiveness on various devices and browsers.

5.1 Hide/Show based on screen width

All components/widgets and Row/Columns can be either marked as hidden or shown based on screen width.

Properties:

1. Hide if screen size is less than or equal to
2. Show if screen size is less than or equal to

Example:

If certain section of screen needs to be hidden on mobile, then one can use "Hide if screen size is less than or equal to" property with appropriate value.

5.2 Wrap/No-wrap of content

By default columns inside any non-Flex Row/Inner-Row will wrap on smaller device. However, if for some reason this needs to be overridden then one can set following Row/Inner-Row property to "No".

Property (For non-flex Rows):

- Wrap content on small screen

For Flex Row/Inner-Row, default behavior is to not to wrap columns. However, it can be controlled through following property.

Property (For flex Rows):

- Flex Wrap

5.3 Context Variables

Following context variables are made available and can be utilized to conditionally set any property values using Rules.

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- CLIENT_BROWSER (Chrome, Edge, Firefox, Safari, Opera, or unknown).
- CLIENT_DEVICE_TYPE (mobile, tablet, desktop, or unknown).
- CLIENT_ORIENTATION (landscape, portrait, or unknown).
- CLIENT_OS (iOS, iPhone, iPad, iPod, android, blackberry, windows, fxos, meego, television, or unknown).
- CLIENT_SCREEN_WIDTH (e.g. 1920, 1366, 1024)

Example:

If "Min Height" needs to be adjusted based on device type one can utilize CLIENT_DEVICE_TYPE to write a rule to set the value conditionally.

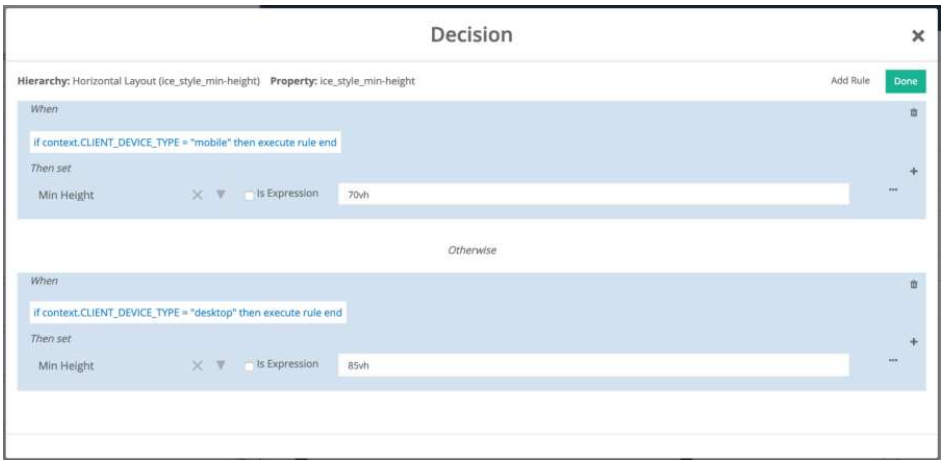


Figure 7: Example of using CLIENT_DEVICE_TYPE to set Property value

These variables can be used to make decision the flow routes as well.

Example:

If entirely different page needs to be shown based on device type.

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Figure 8: Example of using CLIENT_DEVICE_TYPE to conditionally show one page or another

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6 Pitfalls to avoid

6.1 Incorrect usage of "Text" element/component

Digital 1st "Text" element has two parts.

1. Label
2. Display Text

If need is to just display some static text without the need for "Label", please user "Display Text" property of the element and set "Label Position" to "No Label".

- *Do not use "Label" to show static text and leave "Display Text" blank. One will find themselves providing negative margin to pull up the elements below "Text" elements*

6.2 Avoid providing Negative margins

Avoid using negative values for margin. Negative margins are handled different by different browser under certain conditions.

6.3 Setting Image width and height

When using images, either on elements such as Button, Text or using Image element itself; if image width/height needs to be specified then specify both "width" and "height" values. Certain browser calculates "width" based on "height" and some don't. Hence, if either "width" or "height" is specified, one may run into browser compatibility issues.

6.4 Avoid setting Font related properties for Row/Column

Please utilize Theme styles wherever possible and use properties for overrides. Avoid setting style properties (e.g., Font related properties) at Row/Column unless it is required. Properties overrides the theme styles and it will not be changing as theme changes. In addition, it may become hard to debug why theme changes are not reflected.

6.5 Using Flex-Rows/Inner Rows with margin

While using flex rows if requirement is to have space between columns, use "Space Between" property of inner row. Do not use margin/padding to provide space between columns.

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6.6 Remove unused Inner Rows

If there are inner-rows, which does not serve any purpose, please remove.

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7 Testing for different devices

For testing on different devices, one can utilize "Developer Tools" option of chrome browser.

Following steps can be followed:

1. Open developer tools by pressing F12 on windows (Command + option + i on Mac).
2. Click on mobile icon (marked as 1 in the figure below).
3. Choose the appropriate device from dropdown (marked as 2 in the figure below).
4. Enter the URL for application to begin/test.

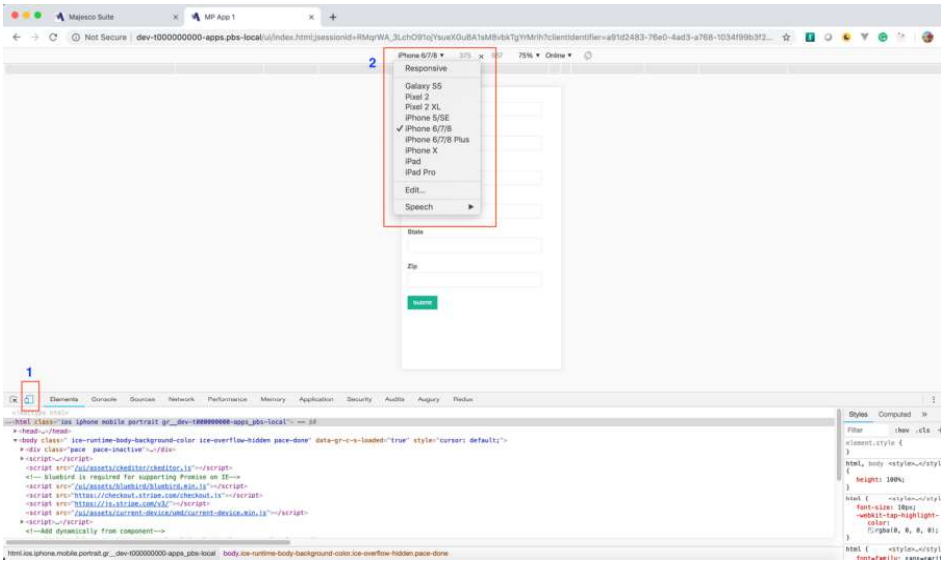


Figure 9: Chrome developer tools

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